ECON2010H

INTERMEDIATE MACROECONOMIC THEORY I

WEEK 09: CONSUMPTION AND SAVINGS: TWO-PERIOD MODEL

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READING: TEXTBOOK CHAPTER 9

OUTLINE

1. Introduction
2. A two-period model
3. The Ricardian equivalence theorem

0. INTRODUCTION

- **Intertemporal decisions** by consumers and firms are at the core of macroeconomics
- Both government policy and monetary policy are intended to influence the **consumption-savings decisions** by consumers and investment by firms
- Government itself may also make intertemporal decisions, regarding government borrowing
- We develop a simple framework with two periods: current period (today) and the future period (tomorrow)
- The key trade-off is the current consumption versus the future consumption
- One important implication of the two-period model is the Ricardian equivalence theorem: tax cut may be ineffective in encouraging current consumption

1. A TWO-PERIOD MODEL

- The model has two periods: current period and the future period, often I may use today and tomorrow
- The model economy consists of consumers and a government
- It is an **endowment economy** with two goods, current consumption and future consumption
    - leaving out production and investment
- The price in the model economy is the real interest rate
- **Perfect foresight**: consumers in current period know their future income, and incomes are certain

1-1. CONSUMERS' PROBLEM AND BEHAVIOR

- The economy has m consumers, $m >> 1$; each consumer lives for two periods
    - Without loss of generality, assume they enter and exit at the same time
    - So that, we do not need to track generations
- All consumers have the same preference of consumption bundles
- No work-leisure decisions, an important omission to keep the model tractable
- Income received by a consumer in each period is exogenous: y in current period and y' in the future period
- Income can differ for different consumers
- Each consumer pays a lump-sum tax t in current period, and a lump-sum tax t' in future period
- In each period, tax is identical for all consumers

1-1-1. BUDGET CONSTRAINTS

- Let c be the consumption level of a consumer in the current period, and s be savings in the current period
- The budget constraint (B.C.) of a consumer in the current period is

$$c + s = y - t \tag{1}$$

- It says that consumption plus savings must equal disposable income
- If $s > 0$ the consumer is a saver/lender, if $s < 0$ the consumer is a borrower

THE CREDIT MARKET

- Where do savers invest and where do borrowers borrow?
- There is a bond market, bonds can be issued by consumers and the government
- Savers buy bonds, borrowers issue and sell bonds
- Two assumptions on the bond market

  1) There is no default risk: bonds are risk-free

  2) Bonds are traded directly, no financial intermediary to facilitate trades

- The real interest rate on bonds is r, **one unit of bond issued in the current period promises to pay $(1+r)$ units of the consumption good in the future period**

- One unit of current consumption exchanges for $(1+r)$ units of the future consumption good

- In other words, the relative price of future consumption in terms of current consumption is $\frac{1}{1+r}$

BUDGET CONSTRAINT IN THE FUTURE PERIOD

- In the future period, disposable income is $y' - t'$, interest and principal on savings is $(1+r)s$

- Budget constraint in the future period is

$$c' = y' - t' + (1+r)s \tag{2}$$

- Consumers spend all resources on consumption in the future period

1-1-2. THE LIFETIME BUDGET CONSTRAINT

- Divide the two sides of budget constraint in the future period by $(1+r)$, then

$$\frac{c'}{1+r} = \frac{1}{1+r}(y' - t') + s$$

- Add current and future budget constraints:

$$c + s + \frac{c'}{1+r} = y - t + \frac{1}{1+r}(y' - t') + s$$

- This leads to the **lifetime budget constraint**

$$c + \frac{c'}{1+r} = y - t + \frac{1}{1+r}(y' - t') \tag{3}$$

- It says, the total consumption (at the price of current consumption good) must equal total disposable income (valued at the price of current consumption good)

REMARKS ON BUDGET CONSTRAINTS

- Consumption c and c' are two different goods, Price of future consumption relative to that of current consumption is $\frac{1}{1+r}$
- All variables are in real terms, no nominal price, no inflation
- No uncertainty and risk:
    - Consumers in the current period know their incomes y' and tax t'
    - Bond market price (interest rate) is risk-free
- No default is necessary and allowed, partly by assuming large enough y and y'

A NUMERICAL EXAMPLE

- $y = 100, y' = 110, t = 20, t' = 25, r = 0.1$
- Then $y + \frac{y'}{1+r} - t - \frac{t'}{1+r} = 100 + \frac{110}{1+0.1} - 20 - \frac{25}{1+0.1} = 157.3$
- The lifetime budget constraint is then

$$c + \frac{c'}{1.1} = 157.3$$

- We follow the textbook to call 157.3 the *lifetime wealth*

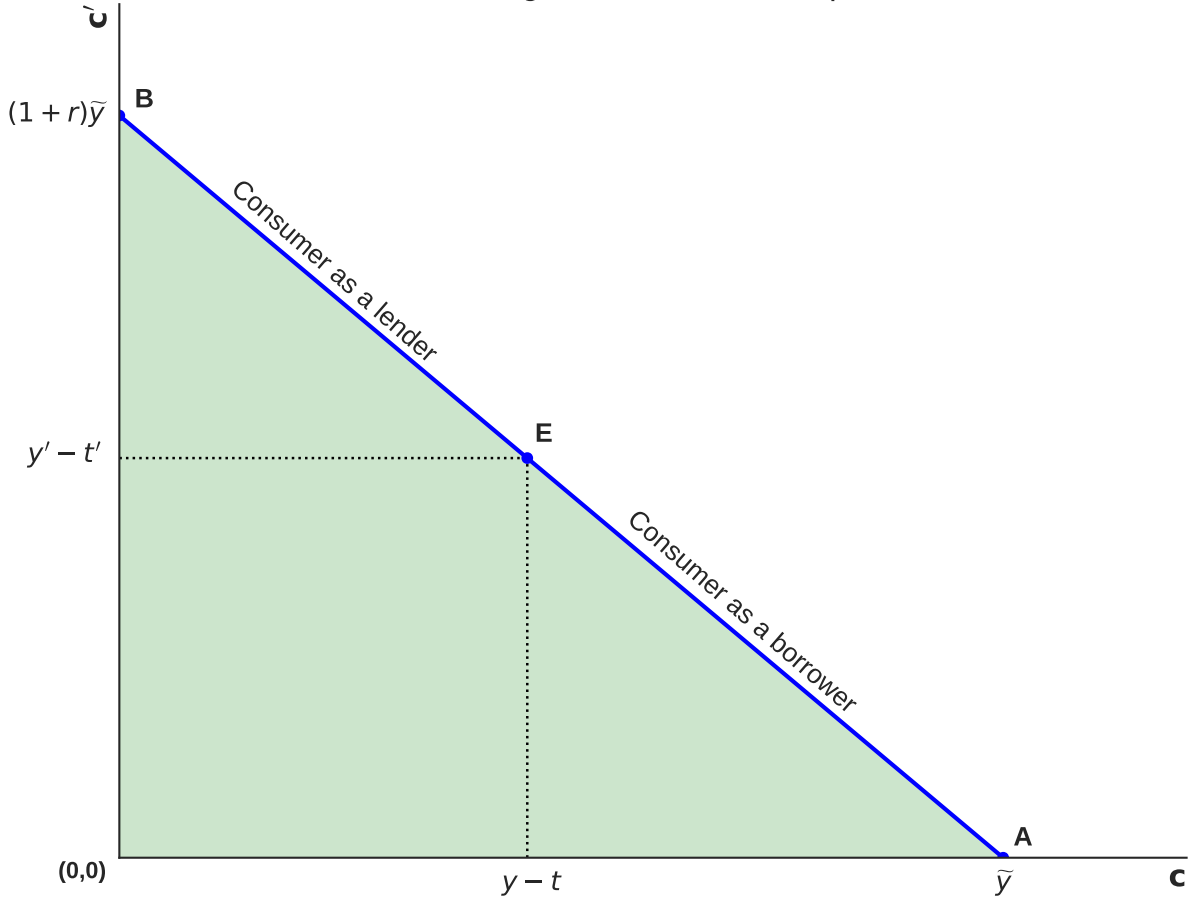
GRAPH OF THE LIFETIME BUDGET CONSTRAINT

- The life-time budget constraint of a consumer is $c + \frac{c'}{1+r} = y - t + \frac{1}{1+r}(y' - t')$
- To ease the notation, use $\tilde{y}$ for a consumer's **lifetime wealth**, thus $\tilde{y} := y - t + \frac{1}{1+r}(y' - t')$
- The lifetime budget constraint can be re-written as

$$(1+r)c + c' = (1+r)\tilde{y} \tag{4}$$

- On graph, to draw the lifetime B.C., first take c' as a function of c, $c' = -(1+r)c + (1+r)\tilde{y}$
    - It is linear, slope = $-(1+r)$
    - Vertical intercept is $c' = (1+r)\tilde{y}$ (by setting $c = 0$)
    - Horizontal distance $c = \tilde{y}$ (by setting $c' = 0$)

The lifetime budget constraint in two-period model



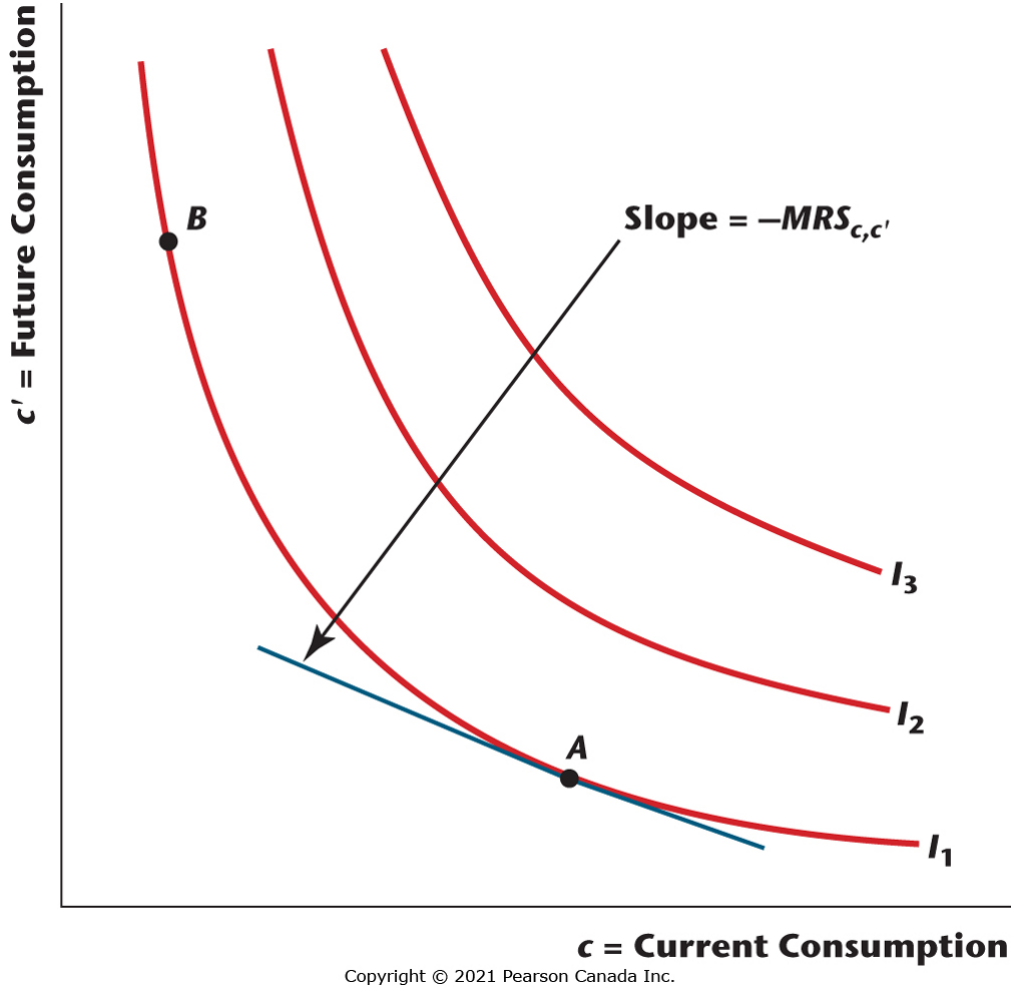
PROPERTIES OF THE LIFETIME BUDGET CONSTRAINT

- Any point in the shaded area including the budget line AEB represents a feasible consumption bundle
- The maximum current consumption is $c = \tilde{y}$, consumer in the current period is unable to consume more than the lifetime wealth
- The maximum future consumption is $c' = (1 + r)\tilde{y}$, consumer in the future period is unable to consume beyond the lifetime wealth (augmented by interest earning)
- Point E: consumer's savings is zero when current consumption equals current-period disposable income, i.e. $c = y - t$
- Segment EB: consumer is a saver/lender as current consumption is below current-period disposable income, i.e. $c < y - t$
- Segment EA: Consumer is a borrower as current consumption exceeds current-period disposable income, i.e. $c > y - t$
- On AEB, for each unit of current consumption given up, the consumer's consumption in the future increases $(1 + r)$ units

1-1-3. CONSUMERS' PREFERENCES

- A consumer receives satisfaction from consumption bundle (c, c')
- A consumer's preference can be represented by a utility function $u(c, c')$
- Preferences here display the same properties as those in the one-period model
    - Non-satiation, more is always preferred to less
    - Consumers like diversity in the consumption bundle
    - Bot current consumption and future consumption are normal goods
- The indifference curve shows consumption bundles leading to the same utility level

- In the two-period model, the property of consumers liking diversity in the consumption bundle says that consumers desire to **smooth consumption** over time
    - Consumers prefer bundle (2,3) to bundle (0,6) and to bundle (5,0)
- The desire for diversity of bundles is equivalent to
    - diminishing marginal rate of substitution of c for c'
    - negative slope of the indifference curve
    - the indifference curve bowed in toward the origin
    - consumption smoothing over time
- On the graph next page, utility level increases when indifference curve shifts from I_1 to I_2 to I_3
- A higher utility value represents a higher level of satisfaction from bundles



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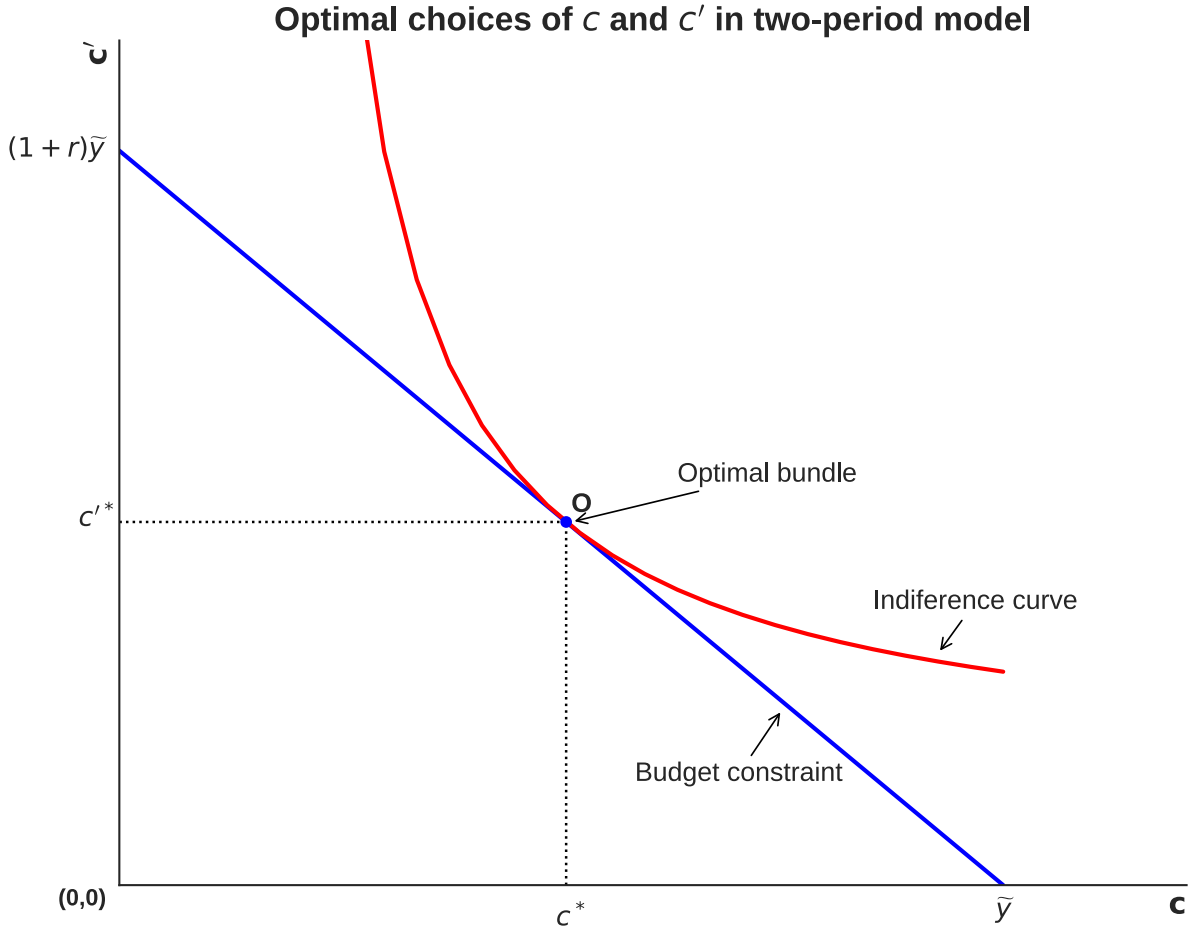
1-1-4. CONSUMERS' OPTIMAL CHOICES

- A consumer's optimization problem is to **maximize** utility $u(c, c')$ (i.e., pushing indifference curve away from the origin as far as possible), **subject to** the lifetime budget constraint $(1+r)c + c' = (1+r)\tilde{y}$
- Solving this optimization problem gives optimal choice of c and c', hence s
- The problem resembles the one-period model in Chapter 4
- The optimal bundle, (c^*, c'^*), is where the lifetime budget constraint is the tangent of the indifference curve
- See the point O on the graph of the next page
- The **optimal savings** is then given by $s^* = y - t - c^*$


Optimal choices of c and c' in two-period model
$\tilde{c}$
$(1+r)\tilde{y}$
c'^*
O
Optimal bundle
Indiference curve
Budget constraint
(0,0)
c^*
$\tilde{y}$
c

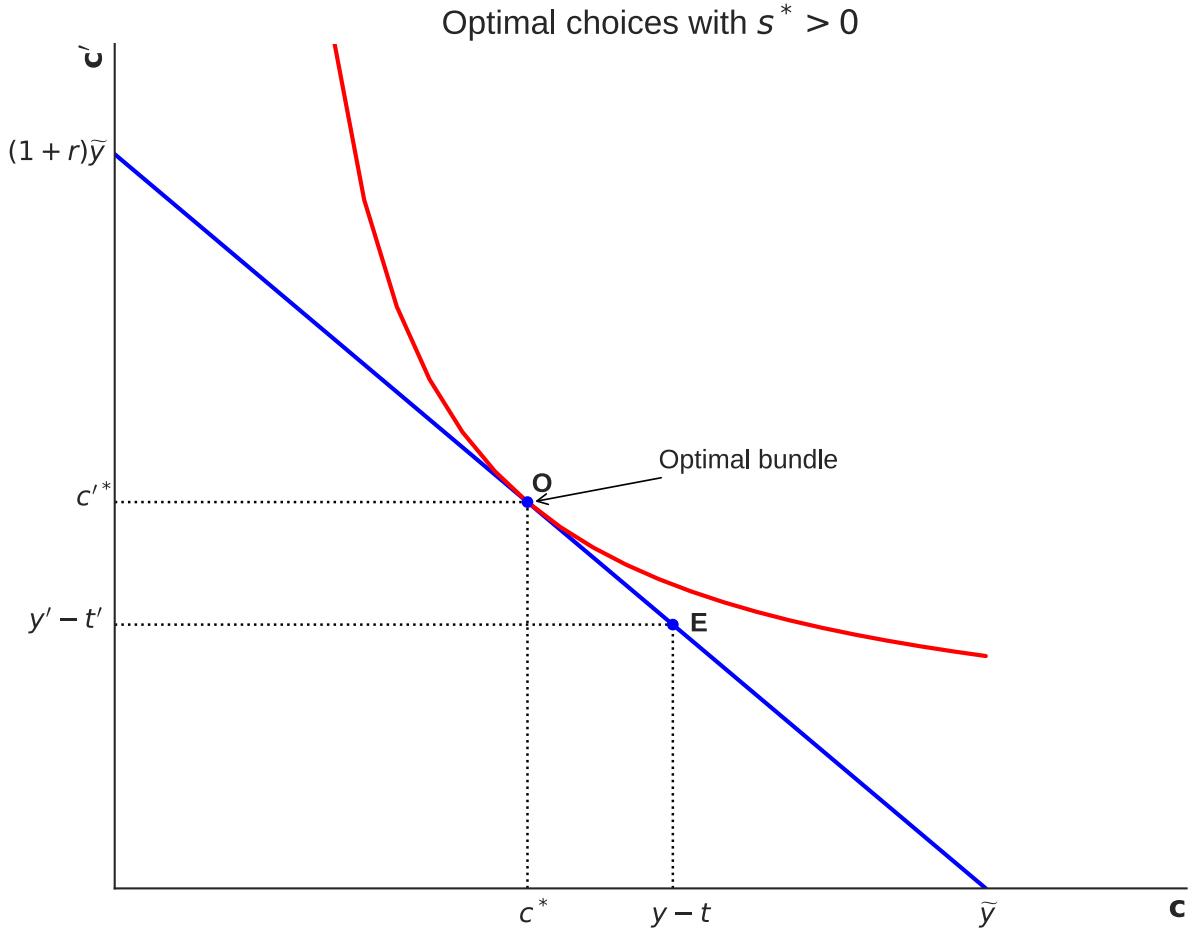
MATHEMATICALLY

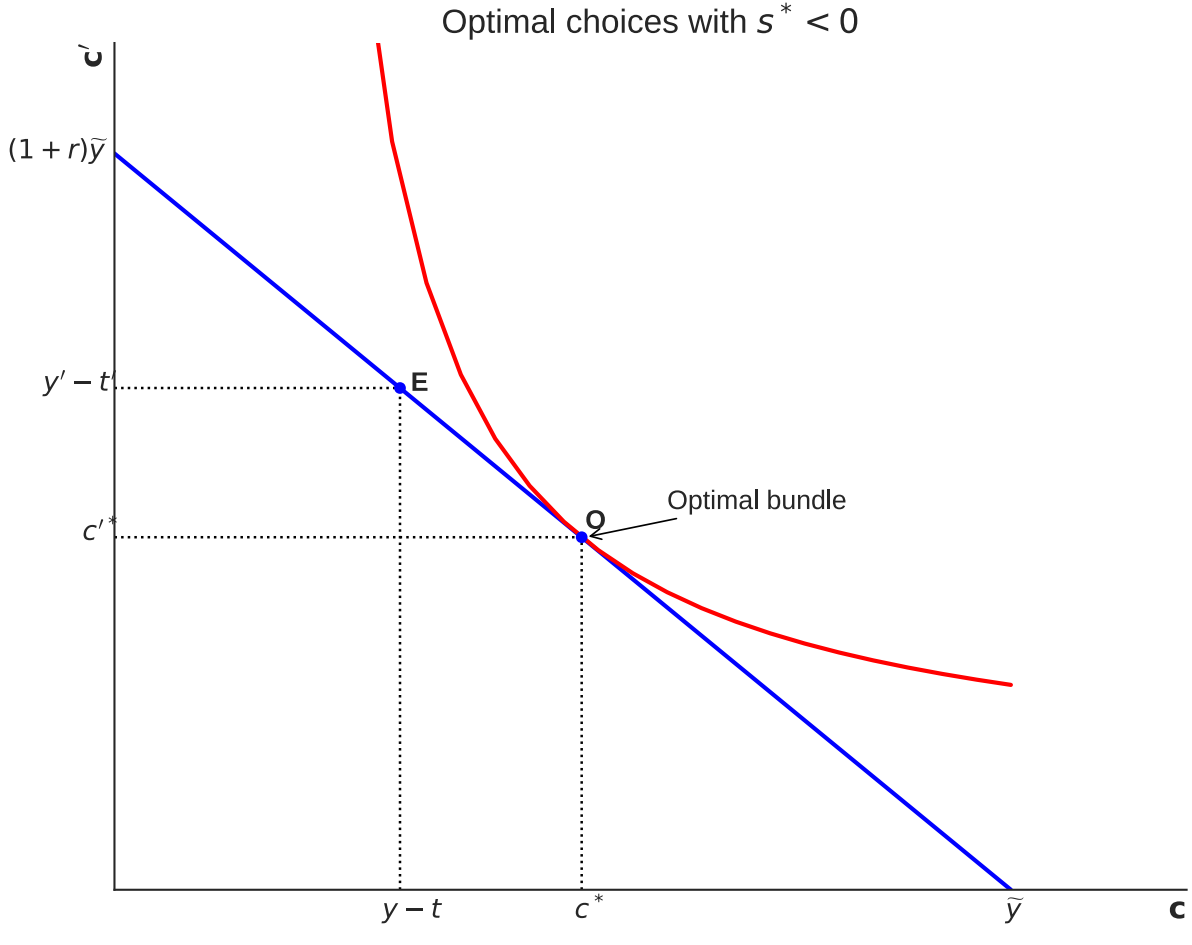
- The optimal bundle shown on the graph suggests that the optimal bundle can be obtained by
    - $MRS_{c,c'} = 1 + r$
    - $c + \frac{c'}{1+r} = \tilde{y}$
- Interpretation: the consumer chooses a bundle such that reducing current consumption by one unit raises the future consumption by $1 + r$ units which is the price of future consumption

- NOTE: the optimal bundle is chosen in the current period

TWO CASES: LENDING AND BORROWING

- If $s^* > 0$, meaning that $c^* < (y - t)$, the consumer is a saver/lender
- This can happen, for example, if current income is (much) higher than income in the future
- If $s^* < 0$, meaning that $c^* > (y - t)$, the consumer is a borrower
- This can happen, for example, if current income is (much) lower than income in the future
- See the graphs of next two slides


Optimal choices with $s^* > 0$
c'
$(1+r)\widetilde{y}$
c'^*
$y' - t'$
O
Optimal bundle
E
c^*
$y - t$
$\widetilde{y}$
c


Optimal choices with $s^* < 0$
$\tilde{c}$
$(1+r)\tilde{y}$
$y' - t'$
E
c'^*
O
Optimal bundle
$y - t$
c^*
$\tilde{y}$
c

A NUMERICAL EXAMPLE

- Suppose that a consumer's preference leads to $MRS_{c,c'} = \frac{c'}{\beta c}$, where $0 < \beta \leq 1$
- The optimal bundle is solved for by
  - the optimality condition $\frac{c'}{\beta c} = 1 + r$
  - the lifetime budget constraint $c + \frac{c'}{1+r} = \tilde{y}$
- From the first condition, $c' = \beta(1+r)c$. Substituting it for c' in budget constraint gives $c + \beta c = \tilde{y}$
- The optimal bundle is $c^* = \frac{1}{1+\beta}\tilde{y} = \frac{1}{1+\beta}\left[y - t + \frac{1}{1+r}(y' - t')\right]$ and $c'^* = \beta(1+r)c^* = \frac{\beta}{1+\beta}[(1+r)(y-t) + y' - t']$

A NUMERICAL EXAMPLE, CONTINUED

- In summary, $c^* = \frac{1}{1+\beta}\left[y - t + \frac{1}{1+r}(y' - t')\right]$;
  $c'^* = \frac{\beta}{1+\beta}[(1+r)(y-t) + y' - t']$
- The optimal savings is $s^* = y - t - c^* = y - t - \frac{1}{1+\beta}\left[y - t + \frac{1}{1+r}(y' - t')\right]$
- Re-arranging it, we have $s^* = \frac{1}{1+\beta}\left[\beta(y-t) - \frac{1}{1+r}(y' - t')\right]$
- When is this consumer a borrower?
- Assigning numbers is unnecessary, nevertheless suppose
  $\beta = 1, y = 2, y = 3, t = 0.4, t' = 0.4, r = 0.05$, then $s^* = -0.44$
- The consumer with these values is a borrower

CHECK YOUR UNDERSTANDING

- The intertemporarl consumption bundle (c, c') treats the same good in different periods as different goods
- When is the optimal savings zero?
- Why is $\frac{1}{1+r}$ the relative price of future consumption?
- How does a change in current tax t affect savings?
- Optional: what is the optimal condition if t is a proportional income tax instead of a lump-sum tax?

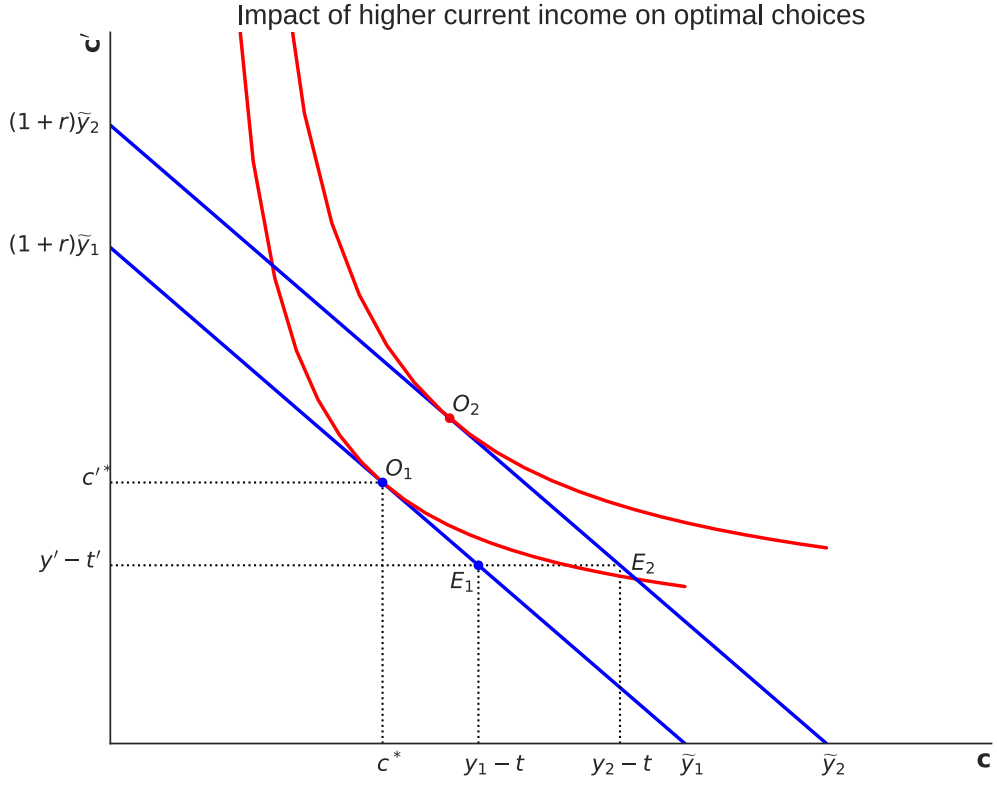
1-2. EFFECTS OF INCOME AND INTEREST RATE ON CONSUMPTION

- Now, go back to the optimal bundle and optimal savings, how does each of the following change optimal choices?
    - An increase in current income
    - An increase in the future income
    - A higher real interest rate
- The effects reverse if income and real interest rate fall

1-2-1. EFFECTS OF A HIGHER CURRENT INCOME

- Suppose current income rises from y_1 to y_2, change by $\Delta y = y_2 - y_1$
- The impact is higher current consumption and higher future consumption
- Savings is also higher: $\Delta s = \Delta y - \Delta t - \Delta c$, here $\Delta t = 0$
- The model predicts **consumption smoothing**: consumers smooth out the effect of a higher current income by changing consumption in both periods
- Consumption smoothing arises from the intertemporal MRS: consumers like diversity
- In data, consumption *expenditure* is volatile for durable goods

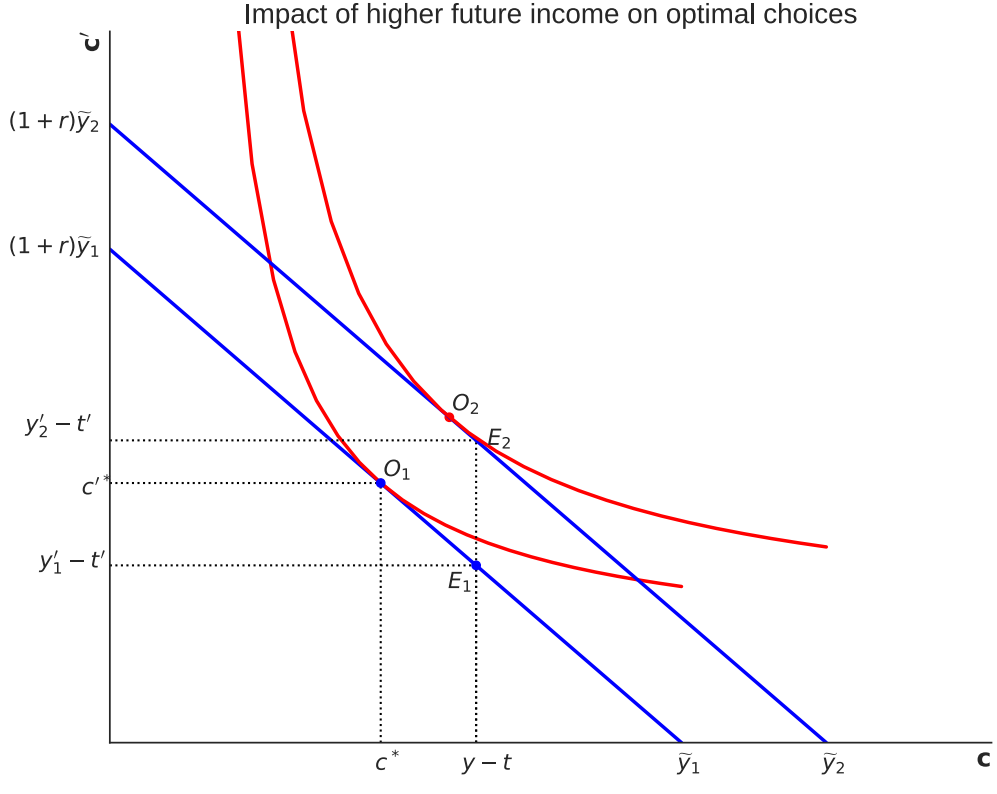
HIGHER CURRENT INCOME SHIFTS THE OPTIMAL BUNDLE FROM O_1 TO O_2



1-2-2. EFFECTS OF A HIGHER FUTURE INCOME

- Future income rises from y_1' to y_2'
- A higher future income also leads to higher consumption in both periods
- But, savings becomes smaller, $\Delta s < 0$
- Consumer raises current consumption, anticipating a higher future income
- **Permanent income** hypothesis: consumption is primarily determined by permanent income
- In the two-period model, permanent income is sum of current and future incomes

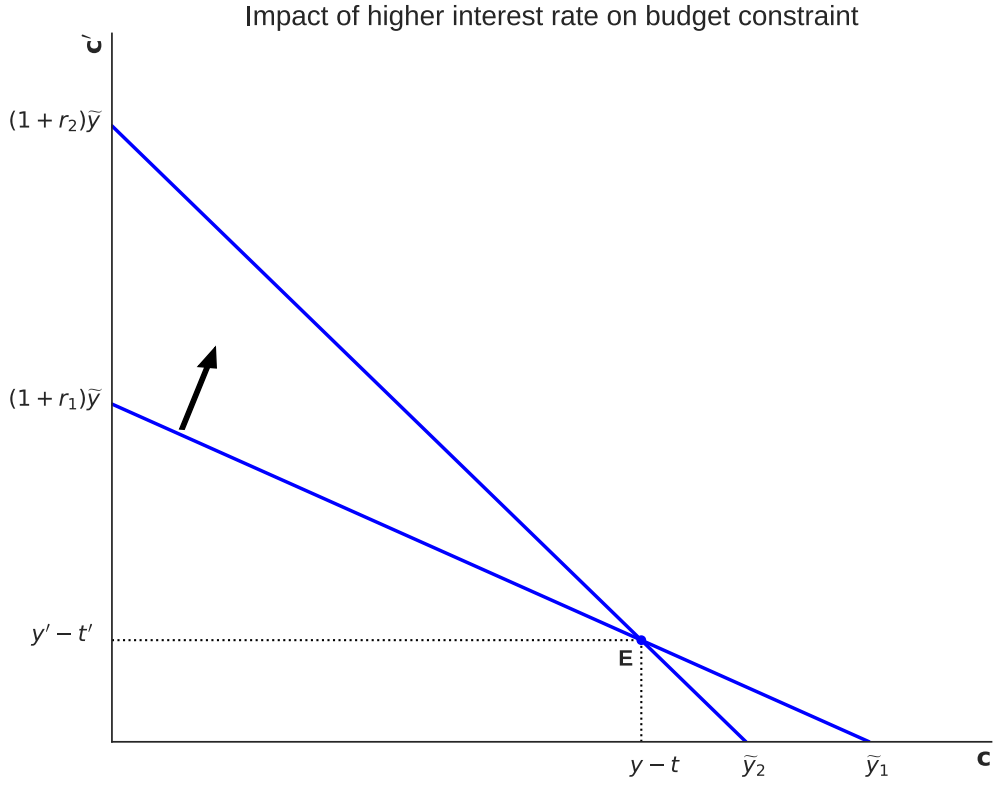
HIGHER FUTURE INCOME SHIFTS OPTIMAL BUNDLE FROM O_1 TO O_2



1-2-3. EFFECTS OF A HIGHER REAL INTEREST RATE

- Suppose real interest rate rises from r_1 to r_2
- A higher interest rate lowers the relative price $\frac{1}{1+r}$, making future consumption cheaper
- Recall, lifetime budget constraint is $(1+r)c + c' = \tilde{y}(1+r)$
- A higher interest rate lowers the life-time wealth since $\tilde{y} = y - t + \frac{1}{1+r}(y' - t')$
- But, $\tilde{y}(1+r)$ is larger, since $\tilde{y}(1+r) = (1+r)(y-t) + (y' - t')$
- Thus, a higher r pivots lifetime budget constraint around the endowment point E

HIGHER REAL INTEREST RATE PIVOTS LIFETIME BUDGET CONSTRAINT

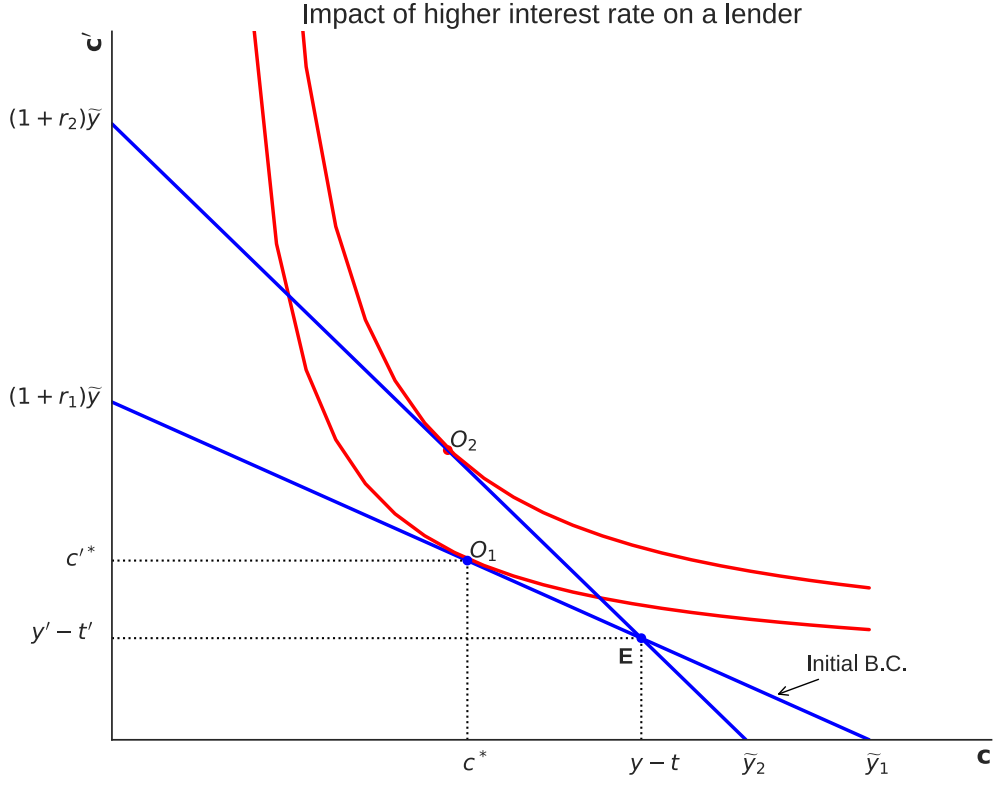


- The effect of a higher real interest rate differs between a lender and a borrower
- If consumer was initially a lender, the optimal bundle was to the left of point E
- If r rises, visually, the utility will be higher for the lender
    - Higher r increases returns to savings
- If consumer was initially a borrower, the optimal bundle was to the right of point E
- Visually, the borrower's utility will be lower as r goes up
    - Higher r increases interest payments, lowers total income

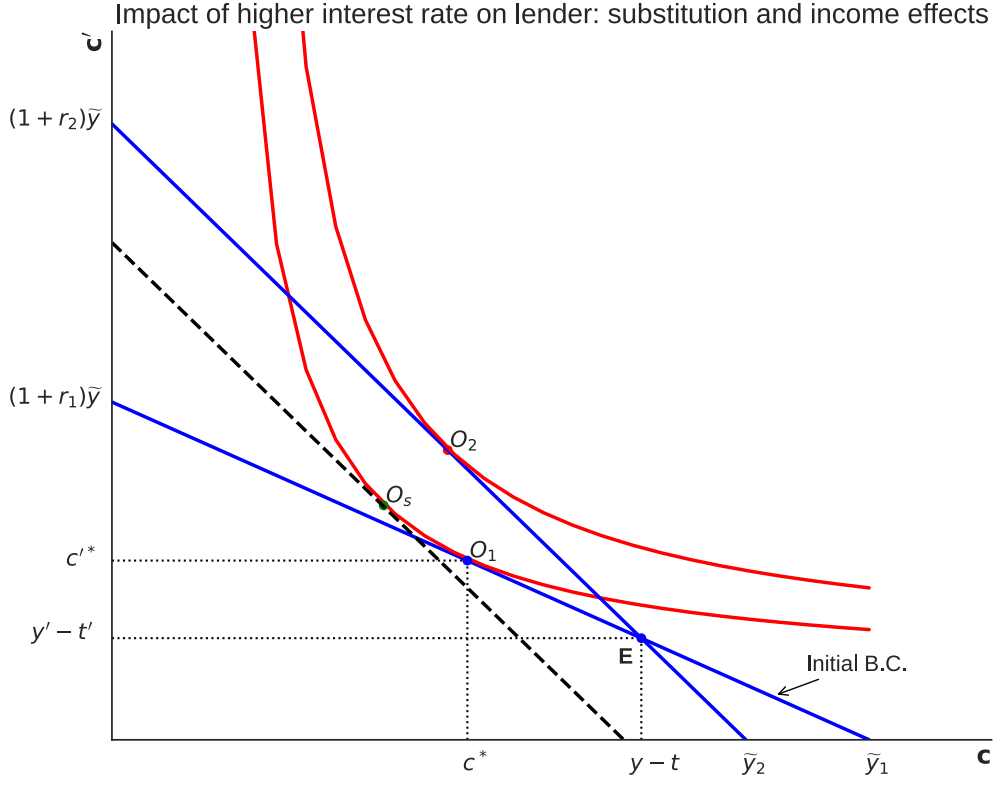
EFFECTS OF A HIGHER REAL INTEREST RATE ON AN INITIAL **LENDER**

- Suppose the optimal savings was positive under r_1: optimal bundle was to the left of endowment point E
- Two effects: the substitution effect and the income effect
- Future consumption increases
    - Substitution effect: higher future consumption as it is relative cheaper as r becomes larger
    - Income effect: total income increases, so does future consumption
- Current consumption may increase or decrease or remains unchanged:
    - Substitution effect: lower current consumption as it is relatively more expensive as r increases
    - Income effect: total income increases, so does current consumption

HIGHER r: LENDER'S OPTIMAL BUNDLE MOVES FROM O_1 TO O_2



HIGHER REAL INTEREST RATE: SUBSTITUTION AND INCOME EFFECTS ON LENDER

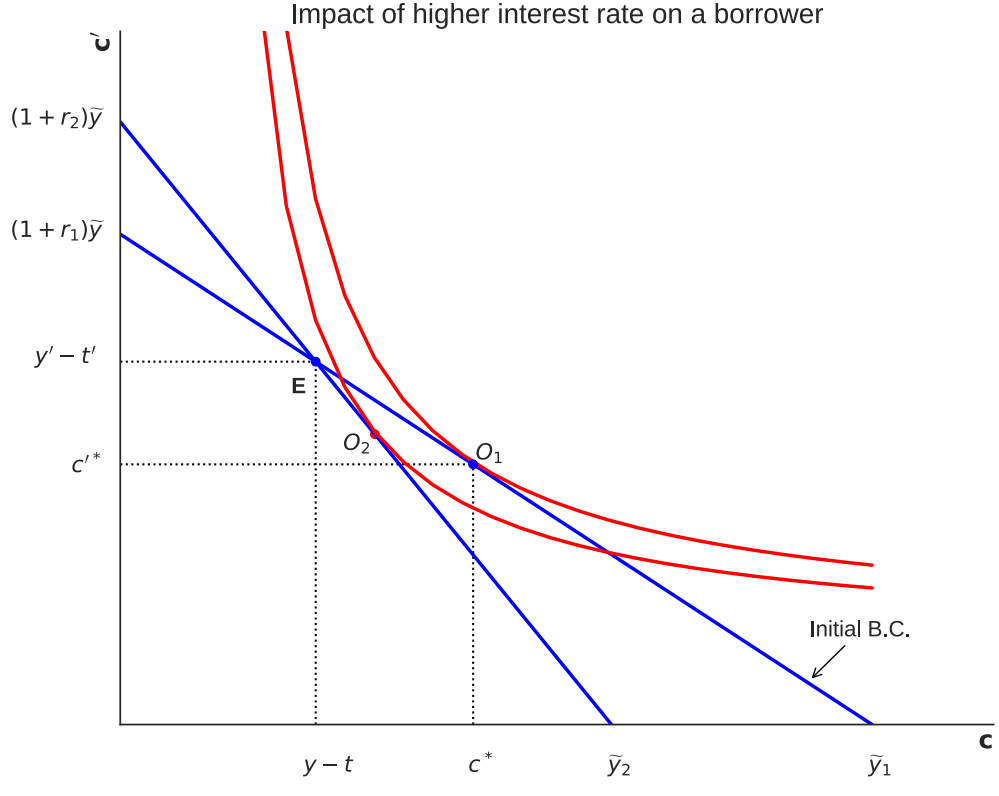


- On the graph of previous slide, change from O_1 to O_s is the substitution effect
- It results in an increase of c' and a decrease of c
- Substitution effect: purely the substitution of c' for c due to a lower relative price, holding utility level being the same as when $r = r_1$
- Change from O_s to O_2 is the income effect
- Income effect: an outward shift of budget constraint, resulting in higher c and c'
- On net, c' increases, c changes ambiguously; thus, savings may go up or down
- Graph shows one possibility, a lower c, savings s becomes larger

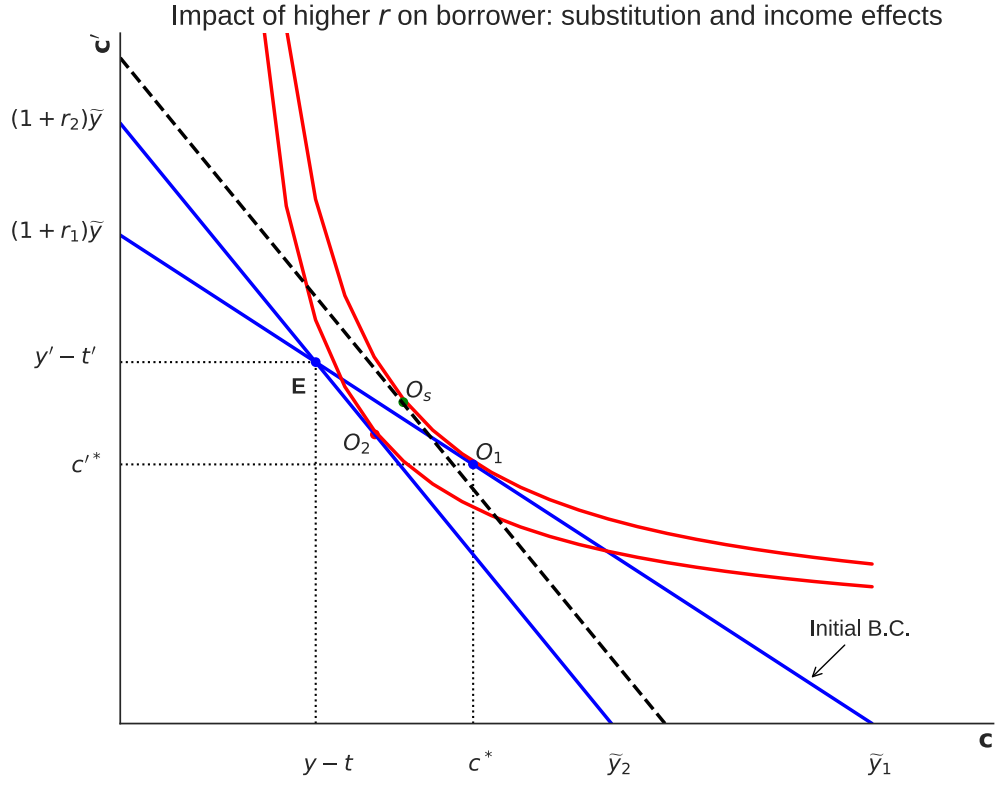
EFFECTS OF A HIGHER REAL INTEREST RATE ON AN INITIAL **BORROWER**

- Suppose the optimal savings was negative under r_1: optimal bundle was to the right of endowment point E
- Pivoted budget line becomes lower following a rise in r
- Two effects: the substitution effect and the income effect
- Future consumption may increase, decrease, or remain unchanged:
    - Substitution effect: higher future consumption as it is relative cheaper as r becomes larger
    - Income effect: total income *decreases*, so does future consumption
- Current consumption falls
    - Substitution effect: lower current consumption as it is relatively more expensive as r increases
    - Income effect: total income *decreases*, so does current consumption

HIGHER r: BORROWER'S OPTIMAL BUNDLE MOVES FROM O_1 TO O_2



HIGHER INTEREST RATE: SUBSTITUTION AND INCOME EFFECTS ON BORROWER



- Substitution effect is shown as a move from bundle O_1 to bundle O_s
    - Lower c and higher c'
- Income effect is shown as a move from bundle O_s to bundle O_2
    - Lower c and lower c'
- Net effect on c' is ambiguous, graphs shows one possibility: higher c'
- net effect on c: lower c
- Implying: higher r reduces borrowing

- In summary: Effects of an increase in real interest rate

Consumer	$\tilde{y}$	**(1+r)$\tilde{y}$**	**Current consumption c**	**Future consumption c'**	**Savings s**
Initial lender	Down	Up	Ambiguous	Up	Ambiguous
Initial borrower	Down	Up	Down	Ambiguous	Up

- Note: for borrower, s is negative, s going up means less borrowing

- Note: For lender, s rises if the substitution effect dominates the income effect

- Numerical example: go back previous example, change r from 0.05 to 0.10, calculate the resulted changes in c, c', s

1-2-4. THE INTERTEMPORAL SUBSTITUTION EFFECT

- In dynamic models like ours, an increase in real interest rate leads to the **intertemporal substitution effect** as we saw above
    - Substituting future consumption c' for current consumption c
- The intertemporal substitution effect implies that overall savings may or may not increase under a higher r
    - Savings by a lender can go up or down
    - Borrowing by a borrower goes down
- The aggregate effect: will a higher r lead to more aggregate savings?
    - It depends on the share of savers and borrowers in the economy
    - It also depends on, among savers, share of those saving more

CHECK YOUR UNDERSTANDING

- When real interest rate rises, Jack puts a larger portion of his income away, is Jack an initial lender or initial borrower?
- Suppose that, in addition to a lump-sum tax, government now imposes a proportional tax on current income but not on future income. After tax income is $(1-\tau)y$.
    - How does the lifetime budget constraint change?
    - How does the proportional tax τ change the optimal bundle?
    - Will consumer savings be higher relative to if $\tau = 0$?

1-3. ANOTHER EXAMPLE: PERFECT COMPLEMENTS

- One type of consumer preferences is the **perfect complements**: consumer prefers a fixed proportion between current and future consumption
- Under this preference, it is always $c' = ac$ with $a > 0$, (regardless of prices)
- It implies, consumers with this preference always have a perfect consumption smoothing
- Finding the optimal bundle needs two equations
    - Preference: $c' = ac$
    - B.C.: $(1+r)c + c' = (1+r)\tilde{y}$

- Solving two equations leads to optimal bundle

$$c^* = \frac{(1+r)\tilde{y}}{1+r+a}$$

and

$$c'^* = ac^*$$

where $\tilde{y} = y - t + \frac{1}{1+r}(y' - t')$

- Optimal savings is $s^* = y - t - c^*$, or

$$s^* = y - t - \frac{(1+r)\tilde{y}}{1+r+a}$$

- Question: will a higher r lead to higher c and c'?
- For an initial lender: c and c' increase, s drops
- For an initial borrower: c and c' fall, s rises (less borrowing)
- Noting, there is not substitution effect

1-4. GOVERNMENT

- For now, assume that government exists also for two periods: current and future
- Assume that government's expenditure G in current period G' in the future period are exogenous
- Total tax revenue is $T = mt$ in current period, and $T' = mt'$ in future period
- Government can borrow by issuing bonds B, so $B < 0$ means government saves
- Assume that government bonds and private bonds bear the same real interest rate r

- Government budget constraint in current period is $G = T + B$
- Government budget constraint in future period is $G' + (1+r)B = T'$
    - Government spending plus total debt payments must equal total tax revenue
    - If $B < 0$, government saves $(1+r)B$ is total income from government lending
- Re-write the B.C. in future period: $B = \frac{T'-G'}{1+r}$
- The **present-value budget constraint** of government is

$$G + \frac{G'}{1+r} = T + \frac{T'}{1+r}$$

1-5. COMPETITIVE EQUILIBRIUM

- The competitive equilibrium is defined as the equilibrium real interest rate r and consumer decisions under this rate

- In the competitive equilibrium, three conditions must satisfy

    1. Consumers decide the optimal levels of c, c', and s, given the real interest rate r
    2. Government's present-value budget constraint holds
    3. The bond market clears

- The last condition is $S^p = B$: the total private savings, S^p, must equal public borrowing

- There is an equilibrium value of real interest rate r under which $S^p = B$

- In equilibrium, bond market clears, $S^p = B$, this means

$$Y = C + G$$

  - where Y is aggregate current income and C is aggregate current consumption
- Why? $S^p = Y - C - T$, $B = G - T$; In equilibrium, $S^p = B$
- Then, $Y - C - T = G - T$

2. THE RICARDIAN EQUIVALENCE THEOREM

- The **Ricardian equivalence theorem states that if current and future government spending are held constant, a change in current taxes with an equal and opposite change in the present value of future taxes leaves the equilibrium real interest rate and consumption unchanged**

- Loosely speaking, a cut in current taxes does not affect consumption and real interest rate, in anticipating that the cut in current taxes is accompanied by a hike in future taxes

- When that happens, we say that the timing of taxes is **neutral** because *a tax cut is not a free lunch*

- The theorem is a strong theoretical result, but empirically is not strong

- The usefulness of the theorem lies in that it makes us think why government deficits in fact matter

2-1. THE RICARDIAN EQUIVALENCE IN THE TWO-PERIOD MODEL

- Recall the government budget constraint $G + \frac{G'}{1+r} = T + \frac{T'}{1+r}$
- Since $T = mt$ and $T' = mt'$, then $G + \frac{G'}{1+r} = mt + \frac{mt'}{1+r}$
- Rearranging it, then $t + \frac{t'}{1+r} = \frac{1}{m}\left(G + \frac{G'}{1+r}\right)$
- Recall, a consumer's life-time budget constraint $c + \frac{c'}{1+r} = y - t + \frac{1}{1+r}(y' - t')$

- Combining government's and consumer's budget constraints, we have

$$c + \frac{c'}{1+r} = y + \frac{y'}{1+r} - \frac{1}{m}\left(G + \frac{G'}{1+r}\right) \tag{5}$$

- Obviously, consumer's budget constraint holds at all time, thus it holds in equilibrium

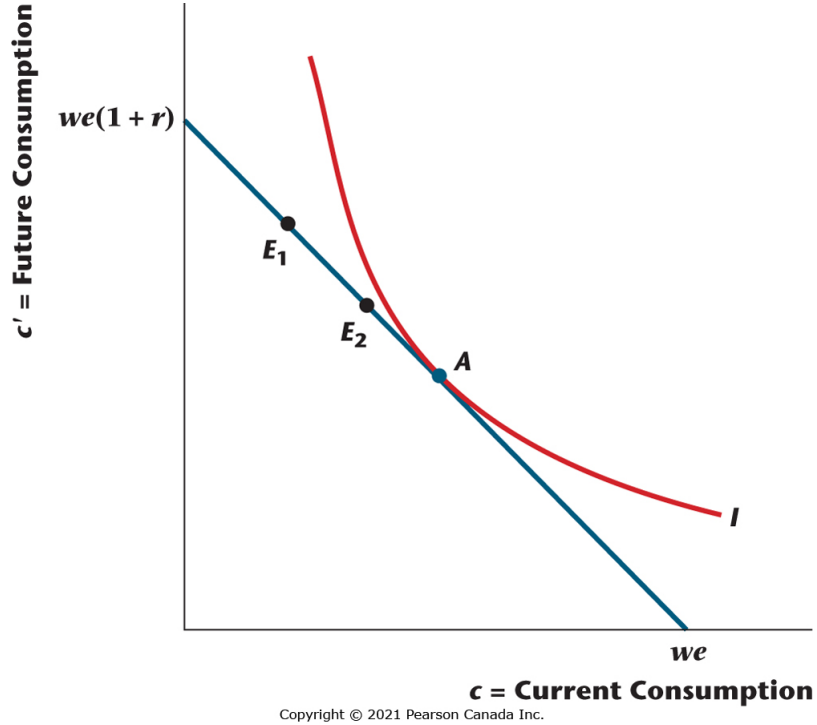
- Suppose that the economy is in equilibrium, all choices are optimal, and $Y = C + G$

- Given $\frac{1}{m}\left(G + \frac{G'}{1+r}\right)$, consumer's budget constraint remains unchanged no matter how t and t' change

- Suppose, for example, government cuts t, budget deficit increases, it must then raise t'

- *Ricardian equivalence*: Since changes in t and t' do not alter the lifetime budget constraint of consumers, these changes do not affect the real interest rate and equilibrium consumption
- The prediction has strong implications for government policy
- According to the Ricardian equivalence, tax cut that is financed by increased budget deficit has no effect on the economy.
- **The timing of taxes has no effect on consumption, but affects private savings**
- $S^p = Y - T - C$. A lower T increases S^p since Y and C are unaffected by a change in T

- Cutting current taxes with a future increase in taxes leaves the lifetime budget constraint unchanged, optimal bundle remains at A
- The endowment point shifts from E_1 to E_2, private savings increases by the amount of the current tax cut



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- Note, we in the graph is $\tilde{y}$

A NUMERICAL EXAMPLE

- $m = 500$,y=10,y'=12,t = 3,t'=4,G = 2000,G' = 1475$
- Then $T = 500 \cdot 3 = 1500$, $T' = 2000$, and $B = 500$
- Suppose that the real interest rate in equilibrium turns out to be $r = 5\%$
- Further suppose that the optimal bundle under 5% interest rate is $c = 6$ and $c' = 9.04$
- Private savings is $S^p = 500 \cdot (10 - 3 - 6) = 500$; then $S^p = B$

- Government's intertemporal budget constraint is $2000 + \frac{1475}{1.05} = 1500 + \frac{2000}{1.05}$
- $Y = 500 \cdot 10 = 5000$, $C = 3000$, thus $Y = C + G$
- Suppose that government reduces taxes in current period to $t = 2$, it then must increase taxes in future period to $t' = 5.05$
- Consumers' lifetime budget constraints remain unchanged
- Lifetime wealth $\tilde{y} = 10 - 2 + \frac{12-5.05}{1.05} = 14.61$ which is the same as before government reduces current taxes
- But, after tax cut, private savings rises to $S^p = 500 \cdot (10 - 2 - 6) = 1000$

2-2. RICARDIAN EQUIVALENCE OFTEN FAILS IN PRACTICE

- In practice, assumptions and the environment differ from the two-period model, making the Ricardian equivalence less empirically relevant
- Redistributional effects of taxes: tax changes affect the wealth of different consumers differently
- Intergenerational redistribution: debt issued by the government today is paid off by future generations
- Taxes are not lump sum; they cause distortions
- Credit market frictions: savers and borrowers receive different interest rates

CHECK YOUR UNDERSTANDING

- Does the Ricardian equivalance theorem suggest that government should not cut current tax but increase current spending?
- If G increases, how will savings change?
- Ricardian equivlance theorem is weak in reality (data), which reason the most important that you think?

RECAP

- Two-period model environment: assumptions and setup
- Budget constraints: in each period, life-time; and intertemporal preferences
- Optimal consumption bundle for both periods is determined in current period
- Who is a lender and who is a borrower?
- Effects on consumption bundle following changes in incomes, and in real interest rate
- Government budget constraints: in each period, life-time
- Competitive equilibrium determines real interest rate with bond market clearing
- Ricardian equivalence